Module 2

Supporting Cabling and Physical Installations

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Learning Objectives

- Summarize Ethernet standards.
- Summarize copper cabling and connector types.
- Summarize fiber optic cabling and connector types.
- Describe physical installation factors for rack-based installations in server rooms and datacenters.
- Deploy and troubleshoot Ethernet cabling.

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Lesson 2.1

Ethernet

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Network Data Transmission Terms

CompTIA

Data transmission is transferring data from one device to another.

Data is sent point-to point via wired or wireless streams or **channels**.

Digital signals (1s and 0s) are transmitted across the channels.

Transmission media include copper wires, fiber optic cables, or wireless signals.

The speed at which the data is sent is the **data transfer rate**.

Error detection/correction mechanisms detect transmission errors.

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Ethernet Standards

IEEE 802.3 standards define the physical layer and data link layer's media access control (MAC) for wired Ethernet.

Standard	Cables	Speed
10Base-T	Unshielded twisted pair (UTP)	10 Mbps
100Base-T	Cat5e or higher	100 Mbps
1000Base-T Gigabit Ethernet	Cat6 or higher	1000 Mbps
10GBase-T 10 Gigabit Ethernet	Cat6 or higher	10 Gbps
100Base-TX Fast Ethernet	Cat5 or higher	100 Mbps

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Carrier Sense Multiple Access with Collision Detection (CSMA/CD)

CompTIA

Purpose

Regulates communication in networks with shared transmission mediums

Process

Carrier Sense

Multiple Access

Collision Detection

If two devices transmit simultaneously, a collision is detected.

Resolution

Jam Signal

Backoff Algorithm

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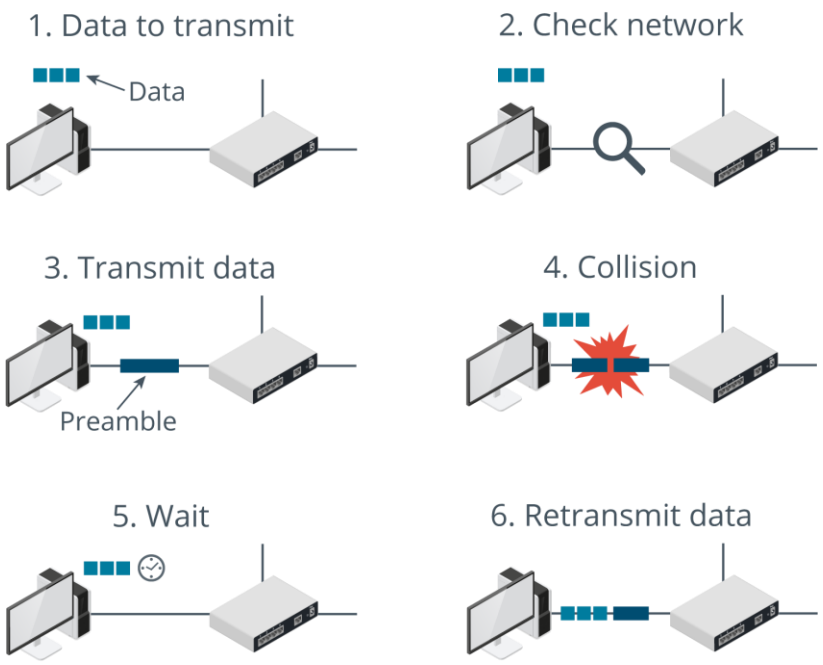
CSMA/CD Diagram



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Fiber Ethernet Standards

Standard	Cables	Speed
100Base-FX	MMF (OM1)	100 Mbps
100Base-SX	MMF (OM1, OM2)	100 Mbps
1000Base-SX	MMF (OM2, OM3)	1 Gbps
1000Base-LX	MMF (OM1, OM2, OM3), SMF (OS1, OS2)	1 Gbps
10GBase-SR	MMF (OM2, OM3, OM4)	10 Gbps
10GBase-LR	SMF (OS1,OS2)	10 Gbps

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Activity: True or False

CSMA/CD regulates communication in networks with shared transmission media.

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Lesson 2.2

Copper Cables and Connectors

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Unshielded Twisted Pair (UTP)

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Structure

- Contains 2 to 1800 pairs in a plastic jacket

Interference

- Minimal EMI protection

Use

- Popular for LANs because it is cost-effective

Categories

- Grades like Cat5e (1 Gbps) and Cat6a (10 Gbps)

Installation

- Avoid sharp bends, keep away from EMI sources

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Shielded Twisted Pair Cable (STP)

CompTIA

- **Structure**
  - Twisted copper wires
  - Conductive shielding
- **Interference**
  - EMI protection
  - Reduced crosstalk
- **Use**
  - High-interference areas
  - Sensitive data transmission
- **Categories**
  - Grades such as Cat5e, 6, 7
- **Installation**
  - Careful handling
  - Avoid sharp bends

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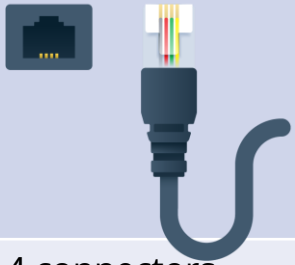
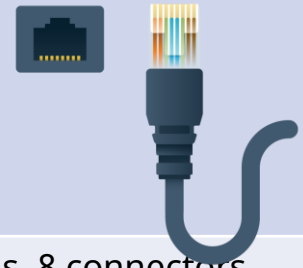
Ethernet Connectors

Feature	RJ11	RJ45
Image		
Configuration	6 positions, 4 connectors	8 positions, 8 connectors
Usage	Phone and modem	Computer networking
Bandwidth	24 Mbps	10 Gbps over Ethernet

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Plenum and Riser-Rated Cable

CompTIA

Plenum-rated

- Installed in plenum spaces
- Made of fire-resistant low toxic materials
- Meets higher fire safety standards
- More expensive

Riser-rated

- Used in vertical spaces between floors
- Fire-resistant
- Prevent spread of fire between floors
- More cost-effective

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Coaxial and Twinaxial Cable

Feature	Coaxial	Twinaxial
Image		
Configuration	Central conductor, insulation, metallic shield, outer jacket	Two inner conductors in a twisted pair, insulation, outer jacket
Usage	TV, Internet, radio signals, CCTV	10 GB Ethernet networks
Bandwidth	Wide range of frequencies, high-speed data transmission	Very-short range high-speed
Connectors	BNC, TNC, SMA	Proprietary

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Activity: Multiple Choice

A company is setting up a network in an industrial environment where machinery often causes significant electromagnetic interference. The network requires a cabling solution that can handle high-speed data transfer while also being resistant to this interference. The cable will be used to connect servers within the same data center, and the runs will not exceed 100 meters.

A. Unshielded Twisted Pair (UTP)
B. Shielded Twisted Pair (STP)
C. Coaxial Cable
D. Fiber Optic Cable

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Lesson 2.3

Wiring Implementation

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Structured Cabling System

A structured cabling scheme is a standard way of provisioning cabled networking for computers in an office building.

Work Area	• User connection point
Horizontal Cabling	• Floor-level networking • Multiple IDFs
Telecommunications Room	• Equipment hub • Dedicated space
Backbone Cabling	• Vertical connections
Entrance Facilities/Demarc	• Network transition point • LEC integration

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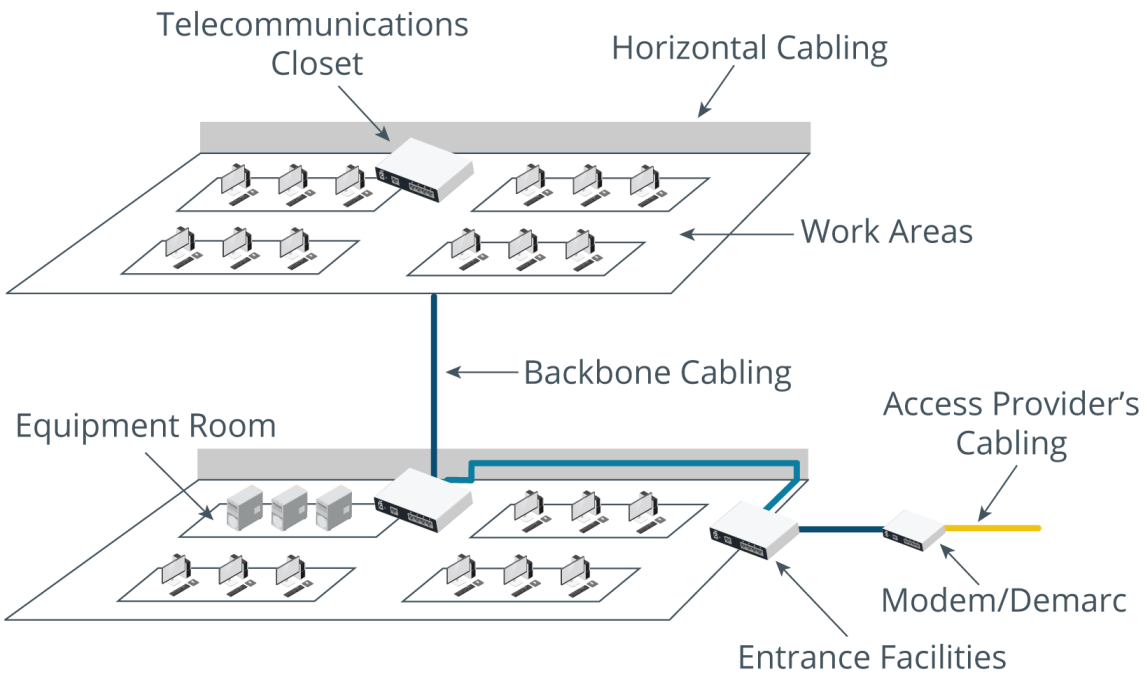
Structured Cabling Diagram



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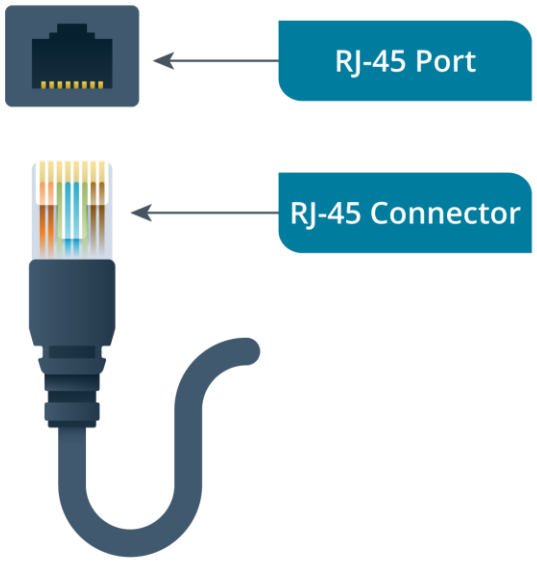
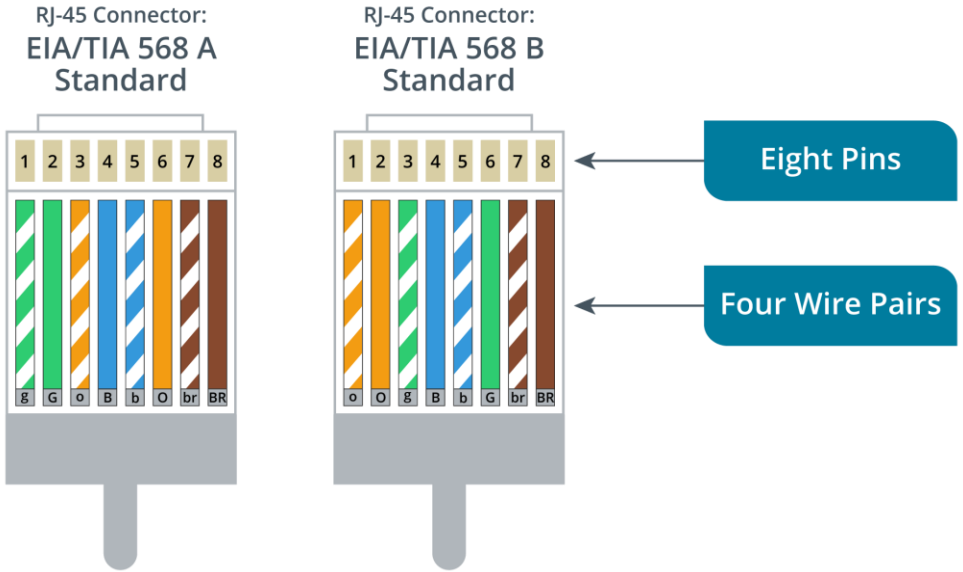
T568A and T568B Termination Standards





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Patch Panels

Purpose	Structure	Function	Wiring Standards
• Facilitates cable management	• Punch down blocks at rear for cable termination • RJ45 ports in front for device connections	• IDC terminals allow for secure and compact cable termination • RJ45 ports enable easy reconfiguration of network connections	• Supports T568A wiring schemes • Supports T568B wiring schemes

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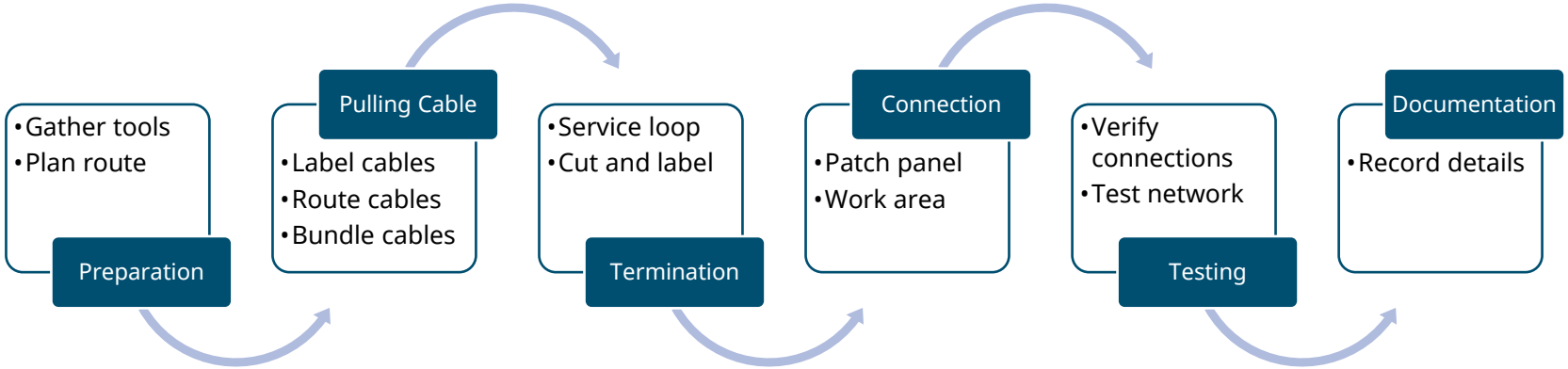
Structured Cable Installation



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Termination Tools

Cable Cutters

For clean cuts of the cable without damaging wires.

Crimp Tools

To attach connectors to the cable ends.

Punch-Down Tools

For terminating cable wires in punch down blocks.

Cable Strippers

For safely removing insulation without nicking wires.

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Activity: What is it?

1

2

3

4

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Lesson 2.4

Fiber Optic Cables and Connectors

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Fiber Optic Considerations

Fiber optic cables

- Thin stands of glass or plastic
- Transmit data using light pulses

Composition

- Core: ultra-pure strands of glass
- Cladding: reflects light back into the core
- Buffer: outer protective jacket

Benefits

- Transmit data up to 800 Gbps
- Maintain signal quality over long distances
- Immune to electromagnetic interference

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Single Mode and Multimode Fiber

Factor	Single Mode	Multimode
Core diameter	Small core allowing only one light to pass through	Larger core allowing multiple light modes to pass through
Light transmission	Light travels directly down the fiber with minimal reflection and attenuation	Light bounces more causing more dispersion and attenuation
Distance and bandwidth	Suitable for long distances	Best for shorter distances
Applications	Long-haul networks, high-speed broadband, telecommunications	Datacenters, LANs
Cost	More expensive	Less expensive

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Fiber Optic Connector Types

Factor	Subscriber Connector (SC)	Lucent Connector (LC)	Straight Tip Connector (ST)
Image			
Ferrule Size	2.5 mm	1.25 mm	2.5 mm
Applications	Telecommunications networks, data transmission	Heavily populated patch panels and cabinets	LANS and datacenters
Features	Push-pull design, quick deployment, low insertion loss, durability	Square shape, duplex header design	Bayonet twist-lock, older but still widely used

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Wavelength Division Multiplexing (WDM)

Allows multiple data channels to share the same fiber using different wavelengths of light

Bidirectional WDM (BiDi):

- Supports Tx and Rx over the same fiber strand
- Uses shifted wavelengths (1310 nm for Tx, 1490 nm for Rx)
- Requires installation in opposite pairs
- Documented in Ethernet standards (1000BASE-BX, 10GBASE-BX)

Coarse WDM (CWDM):

- Supports up to 16 wavelengths
- Typically used for 4-8 bidirectional channels on a single strand
- Can also be used for unidirectional channels on dual strands

Dense WDM (DWDM):

- Provisions more channels (20, 40, 80, or 160)
- Requires precise lasers due to less spacing between channels
- Supports multi-channel 1G, 10G, and 40G Ethernet links
- Transceivers installed in opposite pairs for proper function

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Activity: Two Truths and a Lie

Fiber optic cables are made of thin strands of glass or plastic that transmit data using light pulses

Single mode fiber causes more dispersion and attenuation then multimode fiber

Lucent connectors are often used for heavily populated patch panels

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Lesson 2.5

Physical Installation Factors

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Rack Systems

Secure Areas:

- Telecommunications closets, equipment rooms, server rooms
- Datacenters for server provisioning

Access Control:

- Physical access controls for authorized entry

Rack Installation:

- Steel shelving systems for standard-size equipment
- Secure, compact storage; increased density

Standard Sizing:

- EIA standard 19"/48.26 cm width for rack-mounting

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Humidity and Temperature

Temperature Monitoring	Humidity Control	Electrical Stability	Flood Detection
Prevents overheating	Avoids condensation and corrosion	Ensures uninterrupted power supply	Identifies water-related risks
Ensures effective heat dissipation	Reduces static charge risks	Monitors for voltage irregularities	Triggers immediate system shutdowns

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Power Management

Power Essentials

- Essential stable power for network appliances
- Protection against voltage spikes, surges, and failures

Load and Voltage

- AC circuits aligned with equipment load
- High-voltage circuits in datacenters

PDUs

- Ensure clean power signals
- Offer remote monitoring/control

Battery Backups

- Component-level storage device protection
- UPS for system-level power continuity
- UPS runtime dependent on load/model

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Fire Suppression

Fire Safety Elements

- Clearly marked fire exits.
- Regular emergency evacuation drills.
- Fire-resistant building design.

Detection and Alarms

- Automatic and manual smoke/fire detection systems.

Extinguishers:

- Various types for different fire classes.
- Class C extinguishers for electrical safety.

Sprinkler Systems

- Heat-triggered, water discharge.
- Risk of burst pipes, accidental triggering.

Sprinkler Alternatives

- Dry-pipe
- Pre-action
- Halon (phased out).
- Clean agent

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Activity: Prevention

What would you do to:

- Prevent equipment from overheating in server room
- Prevent static discharge
- Protect equipment and data from brownouts and blackouts

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Lesson 2.6

Cable Troubleshooting

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Specifications and Limitations: Part One

Understanding Specifications	Speed vs. Throughput	Throughput Factors
• Compare expected vs. actual performance • Assess speed, throughput, distance	• Physical Layer: Symbols transmitted, measured in baud rate (Hz) • Data Link Layer: Nominal bit rate or bandwidth (bps)	• Average data transfer rate over time • Affected by encoding, errors, distance, interference

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Specifications and Limitations: Part Two

Measurement Layers	Latency	Distance Limitations
• Network/Transport Layer: Throughput • Application Layer: Goodput (accounting for packet loss)	• Speed of packet delivery, measured in milliseconds (ms) • Also known as latency or delay	• Media type dictates bit rate over distance • Attenuation (dB loss) and interference (SNR) impact performance

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Cable Issues: Symptoms

- Slow Internet speeds
- Random disconnections and reconnections
- Connection timeouts
- Slowing down of applications
- Lagging audio/video communication
- Frequent drops or unstable network link

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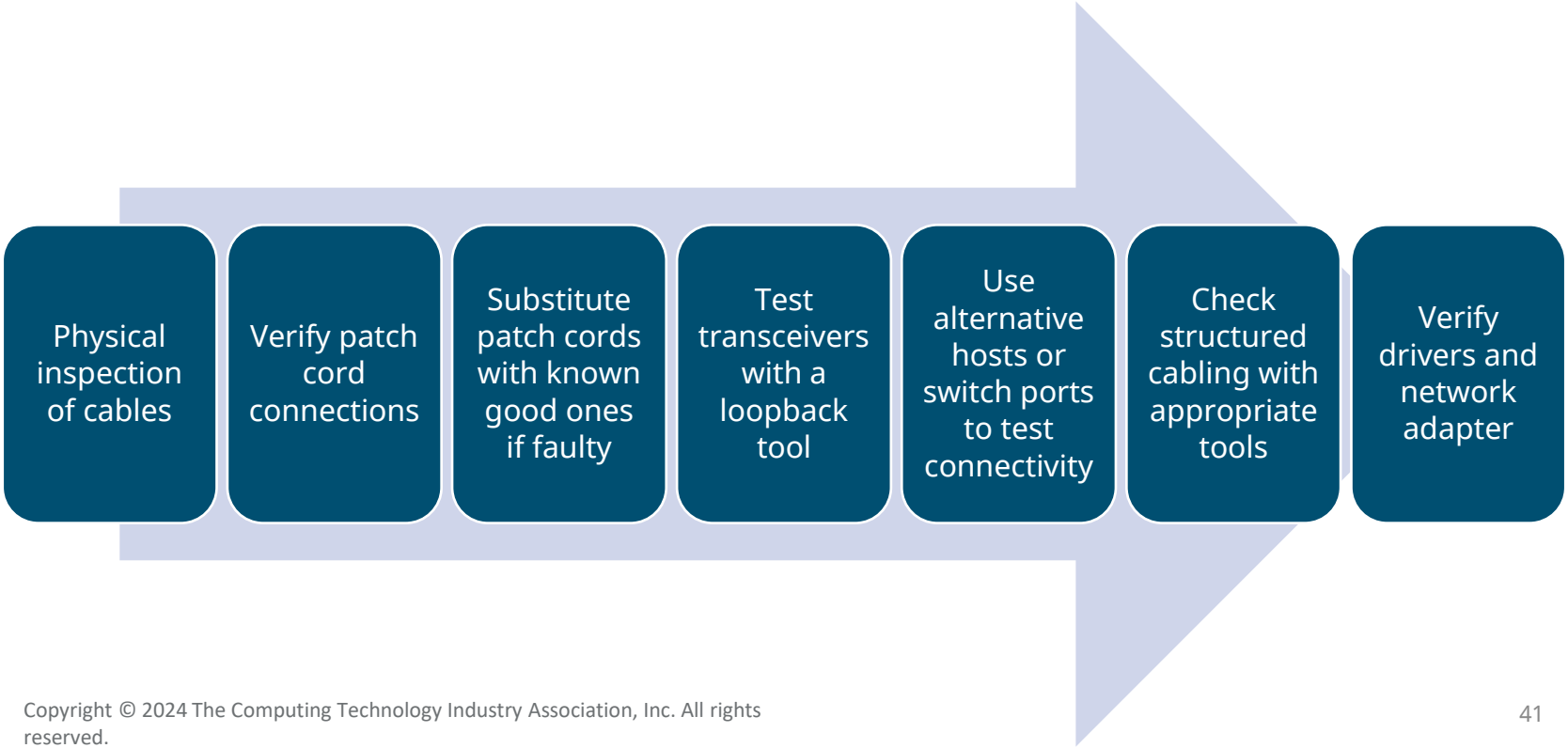
Cable Issues: Troubleshooting



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Cable Testers

Diagnosis with Cable Testing Tools

- Used when cable is not directly accessible
- Diagnose intermittent connectivity or performance issues

Cable Tester Functions

- Reports on physical and electrical properties
- Tests conditions, crosstalk, attenuation, noise, resistance

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Wire Map Testers and Tone Generators

Wire Map Testers

- Detect improper cable termination
- Use base and remote units to test each wire conductor
- Identify issues like open circuits, shorts, and incorrect pin-outs

Tone Generators

- Trace cables through walls or identify active cables in a bundle
- Known as “Fox and Hound”
- Apply signal to trace cable with a probe

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Attenuation Issues

Loss of signal strength in networking cables or connections (measured in decibels (dB) or voltage)

Issues

- Signal strength loss during transmission
- Higher impedance, higher attenuation
- Long cabling distance
- Thin wire size
- Environmental factors

Solutions

- Shorten cable lengths
- Use repeaters/extenders
- Upgrade to high-quality cables
- Measure and test signal attenuation at installation

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Interference Issues

Negative effects of electromagnetic, radio frequency, and electrostatic signals on cable transmissions

Issues

- Electromagnetic interference (EMI)
- Radio frequency interference (RFI)
- Crosstalk from adjacent wires
- Defective connectors/conductors

Solutions

- Shielding techniques (foil/braided shields)
- Systematic cable routing and organization
- Use robust materials and strain relief
- Compatibility testing

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Crosstalk Issues

Interference caused by signal overlap cables

Near End Crosstalk (NEXT)	• Crosstalk at transmitter • Caused by untwisting, faulty shields
Attenuation to Crosstalk Radio, Near End (ACRN)	• Insertion loss vs. NEXT • High value: strong signal
Attenuation to Crosstalk Radio, Far End (ACRF)	• FEXT at recipient end • Independent of link length
Power Sum	• For Gigabit/10 GbE Ethernet • Ensures cable suitability
Alien Crosstalk:	• Interference from nearby cables • Due to tight bundling, bad termination

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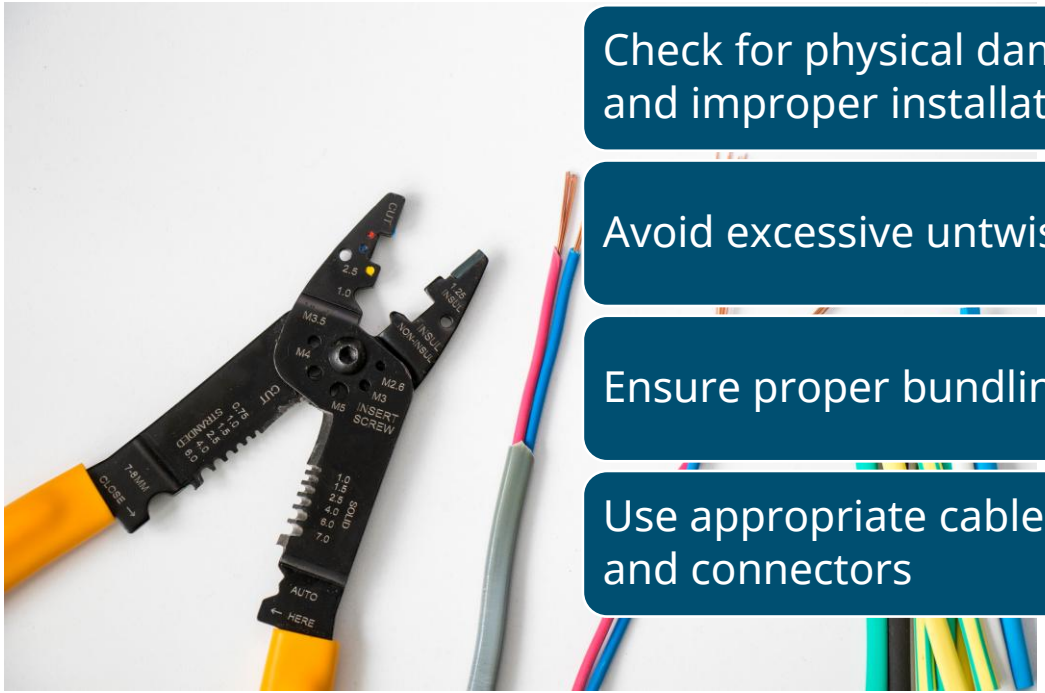
Crosstalk Issues: Solutions



Check for physical damage and improper installation

Avoid excessive untwisting

Ensure proper bundling

Use appropriate cable types and connectors

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Activity: Think About It

What are some symptoms of cabling issues?

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Summary

Understand Cable Needs: Consider factors like interference and attenuation when choosing shielded copper or fiber optic cables

Proper Cable Preparation: Use appropriate tools for cable preparation and termination for connectors or punch down blocks

Application-Specific Use: Ensure cables are used for their intended purpose to meet network requirements.

Cable Testing: Verify cable integrity using appropriate testing tools to identify and troubleshoot faults

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